

Fast reaction of macroinvertebrate communities to stagnation and drought in streams with contrasting nutrient availability

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Abstract: Small, permanent streams are at risk of becoming stagnant or intermittent because of hydrological changes induced by climate change, which can be further intensified by anthropogenic disruptions, such as water abstraction. Macroinvertebrate communities are vulnerable to such changes because they depend on stream hydromorphological regime. We conducted a fully controlled field experiment in 1 impacted and 1 unimpacted Danish lowland stream with contrasting nutrient availability. We used dams and diversions to create short-term (2–10 wk) stagnant and drought conditions, and we installed pools in the drought area to test their value as refugia for benthic macroinvertebrates. After 2 wk, community composition had changed significantly in all treatments in both streams. The abundance of Chironomidae increased and the abundance of mayflies (*Baetis rhodani*), stoneflies (*Amphinemura standfussi*, *Leuctra nigra*), caddisflies (*Silo pallipes*, *Sericostoma personatum*), the amphipod *Gammarus pulex*, and some Diptera taxa (Simuliidae, *Dicranota* sp.) decreased relative to the control. Diversity and total abundance did not change in the stagnant or drought treatments, so we do not consider these variables sensitive to effects of short-term flow reductions. Diversity decreased (unimpacted stream) or was not affected (impacted stream) in pools relative to the drought treatment. Thus, pools did not act as a substantial refugium for macroinvertebrates under extreme low-flow conditions. Current velocity and amount of deposited organic material explained most of the change in the macroinvertebrate community. Nutrient availability did not influence the response of the macroinvertebrate community to the treatments, probably because the physico-chemical changes were exacerbated in the impacted stream and outweighed the expected higher resilience of this community. Our results clearly demonstrate that short-term stagnation and droughts in lowland streams can cause strong alteration of species composition.

Key words: flow reduction, stagnation, drought, water quality, biodiversity

An increase in extreme weather events is expected to take place as a consequence of climate change (Francis and Hengeveld 1998), and this increase will provoke significant changes in flow regimes in lowland streams. Periods with critically low flow will become common in coastal climates of northern Europe during summer (Arnell 1999). Reduction in stream discharge will inevitably affect environmental in-stream conditions by reducing flow velocity, water depth, and wetted perimeter (Dewson et al. 2007b, d, James et al. 2008b) and will increase the amount of deposited material. Furthermore, significant changes may occur in water chemistry and temperature (Cazaubon and Giudicelli 1999, Dewson et al. 2007a, Miller et al. 2007). Anthropogenic disturbance could amplify these changes, especially in areas with high levels of water abstraction (e.g.,

for summer irrigation; Falloon and Betts 2010), thereby increasing the number of temporary streams (Castella et al. 1995).

Benthic macroinvertebrates influence nutrient cycles, primary production, decomposition, and translocation of organic matter. They are an important food source for various fish species (Wallace and Webster 1996) and valuable indicators of the quality of streams (Wilhm and Dorris 1968) because species composition is strongly influenced by in-stream conditions. Flow regime is a key factor for benthic macroinvertebrates in streams because it affects dispersal, habitat use, resource acquisition, competition, and predator–prey interactions (Hart and Finelli 1999). Thus, an altered flow regime has the potential to modify macroinvertebrate performance, species distribution, and abun-

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dance. Reduced discharge can alter macroinvertebrate community composition and diminish biodiversity directly because species with preferences for higher flow are lost (Degani et al. 1993). However, previous studies present mixed conclusions concerning the effect of reduced flow on macroinvertebrates. Some authors reported a decline (Dewson et al. 2007b, c, d, James and Suren 2009, Walters and Post 2011) and others an increase (Dewson et al. 2007a) in abundance. Total biomass of some macroinvertebrate groups increases after flow reduction (Walters and Post 2011), and drought conditions can favor small species with short life cycles, thereby altering the functioning of the macroinvertebrate food web (Ledger et al. 2011, 2013, Woodward et al. 2012).

This discrepancy in observed patterns may, in part, reflect differences in the way various studies were conducted. Most such studies were primarily observational, or investigators manipulated flow at a small scale or in artificial channels (e.g., flumes). Very few investigations have been conducted at a larger scale or in natural streams (McKay and King 2006, Dewson et al. 2007b, Walters and Post 2011). Field experiments under natural conditions incorporate natural, biological, and biogeochemical processes (Carpenter et al. 1995). Therefore, flow manipulations of natural streams are needed to predict effects of altered flow regimes on stream ecosystems.

Our goal was to test experimentally the effects of flow reduction on macroinvertebrate communities in lowland streams. To mimic the drying of a natural stream, we created flow reduction at 2 levels: 1) stagnant (the situation occurring before drought with greatly reduced flow) and 2) drought (no flow and dry stream bed). We hypothesized that: 1) flow reduction established as stagnation (current velocity <5% of control flow) and drought would create significant alterations in macroinvertebrate community composition, diversity, and total abundance because of unfavorable abiotic conditions generated by the experimental flow reduction and 2) that changes in community composition, abundance, and diversity would be more distinct under drought than at stagnant conditions. Pools can act as refugia for macroinvertebrates under drought conditions (Williams 1977, Delucchi 1989, James et al. 2008a), so we also hypothesized that: 3) remnant pools would act as drought refugia for macroinvertebrates. We tested our hypotheses in 2 streams with contrasting land use and nutrient availability because water quality might influence the macroinvertebrate response under low-flow conditions (Dewson et al. 2007c, d, James et al. 2008b, Death et al. 2009).

METHODS

Study sites

We worked in 2 small, lowland streams in central Jutland, Denmark. One was an agriculturally impacted stream predominantly fed by surface runoff (mean summer dis-

charge during the study = 49 L/s), and the other was an unimpacted, groundwater-fed stream (39 L/s). The experimental site in the impacted stream was 41 m asl. It was channelized and surrounded by meadows. Because of intensive agriculture in the catchment, the stream was relatively rich in nutrients (mean NO_3^- concentration = 5.10 mg N/L, range = 5.07–5.14 mg N/L; mean PO_4^{3-} concentration = 26 $\mu\text{g P/L}$, range 11–59 $\mu\text{g P/L}$). The stream bed consisted mainly of sand with small amounts of gravel. Macrophytes were abundant with particularly high coverages of *Berula erecta* (Hudson) Coville, *Callitriche platycarpa* L., and *Myosotis scorpioides* L.

The experimental site in the unimpacted stream was 22 m asl. It was meandering and surrounded by meadows (upper reaches) and scattered trees (lower reaches). The catchment was dominated by forest and extensive grasslands, and the stream was poor in nutrients (mean NO_3^- concentration = 21 $\mu\text{g N/L}$, range = 5–45 $\mu\text{g N/L}$; mean PO_4^{3-} concentration = 9 $\mu\text{g P/L}$, range = 1–13 $\mu\text{g P/L}$). The stream bed consisted mostly of gravel, cobbles, and stones with few sandy areas. Macrophytes were scarce.

Experimental design

During summer 2011, we created bypasses to divert water out of the natural stream channel in both streams. We used bypasses only to remove water from the streams. All experiments were conducted in the streams themselves. In each stream, we assigned 50-m-long reaches upstream from the diversions as control reaches (no manipulation), and we created 3 treatment types (stagnant, drought, pool) downstream from the diversions in a 55-m reach (stagnant and drought each = 27.5 m) in the impacted stream and a 75-m reach (stagnant and drought each = 37.5 m) in the unimpacted stream. The stagnant and drought reaches were separated by a dam, and the stagnant reach was upstream of the drought reach. To mimic the situation occurring prior to drought in naturally drying streams, we reduced discharge in the stagnant reaches to ~5% of the discharge in the control reach and flow to <5% of the control flow. In the drought reach, we reduced discharge further by pumping water running out of the stagnant reach away from the stream channel. The number of available natural pools was insufficient, so we created pools in the drought reach by placing plastic buckets in the stream bed. We buried buckets (diameter = 25 cm) so their edges were level with the stream bed, and we slotted the sides to allow horizontal flux of water and movement of invertebrates. We filled the buckets $\frac{1}{3}$ with substrate, resulting in a water depth of ~10–15 cm.

Sampling

The sampling design followed a Before–After, Control–Impact (BACI) design according to Stewart-Oaten et al. (1986) to avoid site and temporal effects, such as emer-

gence. We sampled the control, stagnant, and drought reaches weekly for 5 wk before installing the diversions as a pre-experimental control ('before' phase of the BACI design) to control for site effects, and biweekly for 10 wk after a 2-wk break while diversions were being established (experimental or 'after phase' of the BACI design). We did not sample pools during the pre-experimental phase because they were not present before the diversion. We compared samples from treated reaches to samples from control reaches to control for temporal effects.

We measured dissolved O₂, temperature, electrical conductivity (EC), and pH in the experimental and control reaches with a multimeter (multi 350i/SET, WTW, Weilheim, Germany) at each sampling date. We also installed a YSI probe (6600V2 multiparameter; Yellow Springs Instruments, Yellow Springs, Ohio) in the control and stagnant reaches to log water O₂ concentrations continuously during the experimental period. We did not install a probe in the drought reaches because of low water levels. We also installed water-level loggers (Micro-Diver Groundwater Data-logger; Schlumberger Water Services, Stavanger, Norway) in the stagnant and drought reaches to monitor water levels during the experimental period.

On every sampling date, we collected 3 macroinvertebrate samples from a randomly selected, predefined transect in each reach (control, stagnant, and drought) and in 3 randomly selected pools (experimental phase). We sampled different transects and pools each time to avoid effects of repeated sampling and disturbance. We minimized spatial dependency as best as possible by taking 3 replicate macroinvertebrate samples with at least a distance of 2 sampling areas between samples within a transect on each sampling date. We collected macroinvertebrate samples from the upper 3 cm of the sediment with a 196-cm² Surber sampler (500- μ m mesh). We preserved the whole sample (sediment and macroinvertebrates) immediately in 96% alcohol. We recorded current velocity (handheld flow meter HFA; Höntzsch Instruments, Waiblingen, Germany), depth, and substrate composition at each macroinvertebrate sampling location and classified substrate composition using the AQEM/STAR substrate size categories (AQEM Consortium 2002). We identified invertebrates in the laboratory to species or genus level, except for Oligochaeta (class), and Chironomidae (subfamily), using the keys by Elliott (1977), Hansen (1987), Lillehammer (1988), Dall and Lindegaard (1995), Edington and Hildrew (1995), Nilsson and Holmen (1995), Nilsson (1996, 1997), Glöer (2002), and Wallace et al. (2003). See Appendix S1 for a full species list including abundances during the 2 phases of the experiment.

Data analysis

We conducted all data analyses in R (version 2.15; R Project for Statistical Computing, Vienna, Austria). We used univariate and multivariate nonparametric tests for statistical analyses with sampling dates as strata to account

for the temporal dependence of the BACI design. Because macroinvertebrate communities differed naturally among streams, we analyzed macroinvertebrate data separately by stream and compared only the strength of the treatment effects relative to the control between streams. Before statistical analysis, we $\ln(x + 1)$ -transformed faunistic data to allow analysis with a linear model (principal response curve [PRC] and redundancy analysis [RDA]; van den Brink and ter Braak 1999) of the abundance data, and physicochemical variables were z-standardized. Physicochemical data were spatially dependent within a treatment reach and, therefore, not repeatedly measured (except when pools were separated from the drought reach by lack of surface-water flow). We compared water-level data between stagnant and drought reaches with a Wilcoxon test.

We calculated macroinvertebrate diversity with the Simpson Index of Diversity (Simpson 1949; *diversity* function of *vegan* package in R; Oksanen et al. 2011). We ran a nonparametric test (based on asymptotic null distributions and stratified for sampling date, *oneway_test* function of *coin* package; Hothorn et al. 2006, 2008) for each stream independently for both the pre-experimental and experimental phases to test for treatment effects. If the outcome was significant, we tested which treatments were responsible for the effect by comparing diversity-index values and total macroinvertebrate abundance between treatment and control with nonparametric pairwise tests (*oneway_test*). We used sequential Bonferroni corrections after the pairwise tests to correct α .

We used distance-based (d/b) RDA (*capscale* in *vegan*) based on Bray–Curtis similarity to analyze the relationship between macroinvertebrate community composition and physicochemical variables (all sampling dates). We removed rare species (taxa <1% of total abundance) from the data set prior to analysis. We tested significance of d/b RDAs with permutational analyses of variance (PERMANOVA with 1099 iterations; *anova.cca* in *vegan*). We used variance inflation factors (*vif.cca* in *vegan*) to test explanatory variables for collinearity. Variance inflation factors >10 occurred for the impacted stream (Borcard et al. 2011), so we used forward selection (5000 iterations; *ordiRstep* in *vegan*) to select 7 explanatory variables.

We used PRC analyses based on Euclidean distances (*prc* in *vegan*; van den Brink and ter Braak 1999) to investigate treatment effects over time on physicochemical data (van den Brink and ter Braak 1999). PRC is based on redundancy analysis (RDA) with the scores of the 1st principal component of the treatments plotted over time relative to the RDA scores of the control to show a community PRC for each treatment (van den Brink and ter Braak 1999). We calculated 1 PRC for substrate composition data and 1 for physicochemical variables. We calculated PRCs on faunistic data based on Bray–Curtis similarity, with treatments as factor levels and sampling dates as strata (C. J. F. ter Braak, Wageningen University, personal communication) to

account for temporal dependence of samples taken on different dates. The 1st d/b RDA axis was used for the PRC, which significantly explained the variation in the macroinvertebrate data (PERMANOVA, $p < 0.001$).

We used PERMANOVAs stratified by sampling date to account for the repeated measures design as significance tests for the PRC model. We tested significance of treatment effects in PRCs (pre-experimental and experimental phases separately) with permutational multivariate analyses of variance (permMANOVA with 1099 iterations; *adonis* in *vegan*). The permMANOVAs were based on the same data and similarity indices and used the same models as the PRCs. Thus, significant treatment effects in the permMANOVAs indicate significance in the PRCs. We checked data for homogeneity of variance–covariance matrices (*betadisper* in *vegan*, Borcard et al. 2011) before conducting the permMANOVAs. When the global permMANOVA was significant, we used pairwise PERMANOVAs (sampling dates used as strata) to compare treatments and applied sequential Bonferroni corrections to α . We compared treatment PRC scores between streams (contrasting nutrient availability) with a nonparametric test (*oneway-test*) with sampling dates as strata to test whether the treatment effect was stream dependent.

RESULTS

Treatment conditions

We succeeded in establishing different treatment conditions in our experiment. In the unimpacted stream (primarily groundwater fed), current velocity was consistently <5% of the control flow in the stagnant (range: 0–0.01 m/s) and drought reaches (0–0.02 m/s). Transects used for macroinvertebrate sampling were dry for a significantly longer time in the drought (54.2% of the time on average) than in the stagnant treatment (1.3%; Wilcoxon test, $p = 0.005$). In the impacted stream, current velocity was <5% of the control flow in the stagnant (range: 0–0.06 m/s) and drought (0–0.02 m/s) reaches. However, on 2 sampling occasions, mean flow in the stagnant and drought reaches (0.02 and 0.04 m/s, or 0.01 and 0.01 m/s respectively) reached 9.8 and 18.6% or 6.6 and 5.7% of the control flow, respectively, because of extreme precipitation events. Discharge was higher than desired in the drought reaches of both streams on most dates because of precipitation. The higher flow was the reason that pools were separated from the drought treatment by lack of surface flow on only 1 sampling occasion in the impacted stream and on 2 sampling occasions in the unimpacted stream. Nevertheless, the transects used for macroinvertebrate sampling in the drought treatment were dry for a significantly longer time (1.7%) than the transects used in the stagnant treatment (0%, Wilcoxon test, $p = 0.02$), although the difference was not as pronounced as in the unimpacted stream.

Dissolved O₂ was affected by flow manipulations. In the impacted stream, the mean O₂ concentration calculated from the continuous measurements was 9.19 mg/L (range: 5.38–12.53 mg/L) in the control reach and 8.63 mg/L (0.46–12.47 mg/L) in the stagnant reach. In the unimpacted stream, the mean O₂ concentration was 11.26 mg/L (9.61–12.11 mg/L) in the control reach and 11.05 mg/L (1.21–12.78 mg/L) in the stagnant reach. In the drought treatment, only point measurements were taken. The mean O₂ concentration based on these point measurements was 6.56 ± 1.68 mg/L in the impacted and 10.22 ± 2.32 mg/L in the unimpacted stream.

Treatment effects on abiotic variables

Substrate composition did not differ between treatments and the control during the pre-experimental phase in either stream (permMANOVA, $p > 0.05$). In the impacted stream, a significant change in substrate composition occurred in the drought and pool (permMANOVA, $p < 0.05$), but not in the stagnant treatment (permMANOVA, $p = 0.565$). In both streams, the amount of sand declined, whereas the amount of organic material (mud, coarse particulate matter, leaves) increased in all treatments compared to the control (Fig. 1A, B), and in the impacted stream the amount of small gravel also increased (Fig. 1A). PRC models were significant for both streams (PERMANOVA, $p < 0.005$). We were unable to run a permMANOVA with substrate composition data for the unimpacted stream because the variance–covariance matrices were not homogeneous (Borcard et al. 2011).

Physicochemical variables did not differ between control and treatment reaches during the pre-experimental phase in either stream (permMANOVA, $p > 0.05$; Fig. 1C, D), whereas during the experimental phase significant differences emerged for the PRC model (PERMANOVA, $p < 0.005$) and for treatment effects (permMANOVA, $p < 0.007$). Stagnant and drought conditions resulted in higher temperatures and lower pH (impacted stream only), and lower current velocities, lower dissolved O₂, lower conductivities, and lower water depths compared to the controls (Fig. 1C, D). In the unimpacted stream, variance–covariance matrices were not homogeneous, so no permMANOVA could be conducted, but the PRC model for physicochemical variables was significant (PERMANOVA, $p < 0.005$), and the PRCs (Fig. 1C, D) suggest that removal of water resulted in clear differences in the physicochemical variables and substrate composition of all treatments compared to the control.

Treatment effects on macroinvertebrate diversity and abundance

Simpson's Index (D) did not differ between 2 streams during the pre-experimental phase (impacted: $D = 0.65$,

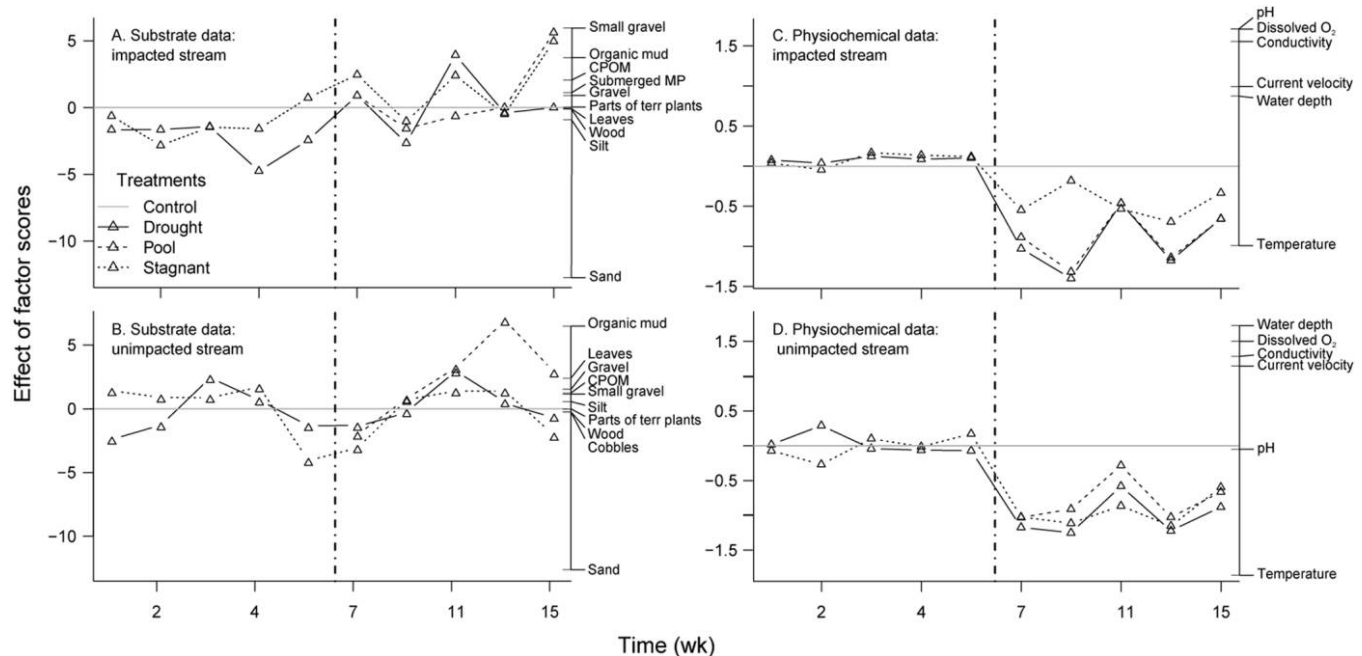


Figure 1. Principal response curves (PRCs) for substrate (A, B) and physicochemical (C, D) data from the impacted (A, C) and unimpacted (B, D) stream. Each point represents the mean of 3 replicates per sampling point. The vertical broken line indicates the beginning of the experimental phase. PRCs show the multivariate response of measured variables within treatments (y -axis) relative to the control (0 line) over time (x -axis). Thus, the effects of physicochemical and substrate factors are displayed within treatments compared to the control on each sampling date over the course of the experiment. The variables with the highest and lowest factor scores (furthest away from the control line in both the positive and negative direction) contribute most to the difference of treatments from the control. CPOM = coarse particulate organic matter, MP = macrophytes, terr = terrestrial.

unimpacted: $D = 0.70$). Flow manipulations did not affect D in the impacted stream. In the unimpacted stream, D was significantly lower in pools than in the control and the stagnant and drought treatments (nonparametric test; Table 1, Fig. 2A, B). In the impacted stream, macroinvertebrate abundance was significantly lower in the drought and pool treatments than in the control and stagnant treatment (nonparametric test; Table 1, Fig. 2C, D).

Treatment effects on macroinvertebrate community composition

In the pre-experimental phase, macroinvertebrate community composition did not differ between control and treatment reaches (permMANOVA, $p > 0.05$), but significant differences between control and treatment reaches emerged in response to flow manipulations during the experimental phase (PERMANOVA, $p < 0.005$) and among treatments (permMANOVA, $p < 0.001$; Table 2). In the impacted stream, the treatment effect was comparable in the pool and drought treatments and less pronounced in the stagnant treatment (Fig. 3A). In the unimpacted stream, the treatment effect was most pronounced in the pool treatment and least pronounced in the stagnant treatment (Fig. 3B). We evaluated the effects of water quality on macroinvertebrate community response to flow manipulations

by using PRC scores to compare observed treatment effects among streams, and we found no difference (nonparametric test, $p > 0.05$; Table 3).

Three subfamilies of Chironomidae (Tanypodinae, Chironomini, and Tanytarsini) responded positively (increase in abundance and occurrence compared to the control) to flow manipulations, whereas *Baetis rhodani* and *Dicranota* sp. responded negatively to these manipulations. Simuliidae and *Gammarus pulex* were negatively affected (decrease in abundance and occurrence in fewer samples compared to the control) by flow manipulations in the impacted stream, and *Leuctra nigra*, *Silo pallipes*, and Orthocladinae were negatively affected in the unimpacted stream (Fig. 3A, B).

Relationship between abiotic factors and macroinvertebrate community composition

Community composition changed significantly in both streams in response to changes in abiotic conditions (d/b RDA; PERMANOVA, $p < 0.001$; Fig. 4A, B). Seventeen percent (adjusted R^2) of the variance in the fauna in the impacted stream and 19% in the unimpacted stream could be explained by the treatments. The first 4 axes of the d/b RDA in the impacted stream and the first 5 axes of the d/b RDA in the unimpacted stream contributed significantly

Table 1. Pairwise comparison tests for significant treatment effects on Simpson's Index of Diversity and total abundance. Pairwise comparisons are shown only if the overall nonparametric test was significant. * indicates significance after sequential Bonferroni correction.

Stream	Variable	Experimental phase	Treatment 1	Treatment 2	<i>p</i>
Unimpacted	Simpson's Index	Experimental	Control	Pool	0.004*
				Drought	0.267
				Stagnant	0.645
			Drought	Stagnant	0.739
				Pool	0.006*
				Stagnant	0.007*
Impacted	Abundance	Pre-experimental	Control	Drought	0.020
				Stagnant	0.060
				Stagnant	0.146
		Experimental	Control	Drought	0.006*
				Pool	0.008*
				Stagnant	0.734
			Drought	Stagnant	0.001*
				Pool	0.989
				Stagnant	0.004*

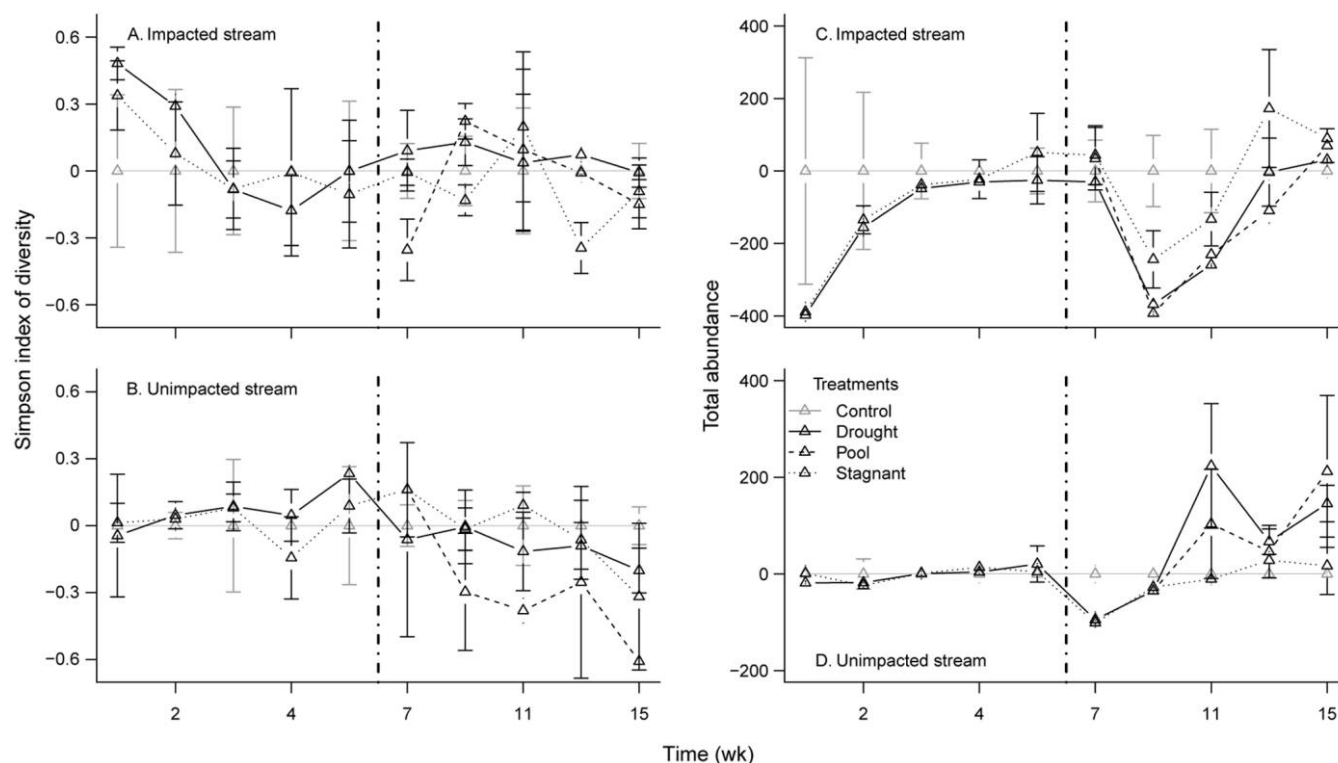


Figure 2. Mean (± 1 SD) deviation of treatment values from control for Simpson's Index of Diversity (A, B) and total macroinvertebrate abundance (C, D) in the impacted (A, C) and unimpacted (B, D) stream. The vertical broken line indicates the beginning of the experimental phase.

Table 2. Pairwise comparison for the treatment effect on macroinvertebrate community composition based on pairwise permutational multivariate analysis of variance. * indicates significance.

Stream	Treatment 1	Treatment 2	R^2	p
Impacted	Control	Stagnant	0.19	0.0002*
		Drought	0.27	0.0002*
		Pool	0.25	0.0002*
	Drought	Stagnant	0.12	0.0002*
		Pool	0.03	0.3084
		Stagnant	0.15	0.0002*
Unimpacted	Control	Stagnant	0.23	0.0002*
		Drought	0.22	0.0002*
		Pool	0.30	0.0002*
	Drought	Stagnant	0.18	0.0002*
		Pool	0.11	0.0002*
		Stagnant	0.24	0.0002*

to explain community composition (PERMANOVAs, both $p < 0.05$). However, the 1st and 2nd axis together explained 11 of 17% and 11 of 19% of the variance in community composition in the impacted and unimpacted streams, respectively, so we used only these axes in the analyses. All explanatory variables used for the d/b RDAs are listed in Table 4, but we used only variables that significantly explained macroinvertebrate composition when interpreting d/b RDAs.

In both streams, low scores on the 1st d/b RDA axis (RDA 1) corresponded with relatively high values of current velocity, conductivity, and % small gravel (Fig. 4A, B), similar to the pre-experimental phase and control reaches in the experimental phase. Invertebrate taxa related to these environmental conditions were *B. rhodani*, *Amphinemura standfussi*, *Dicranota* sp., and Simuliidae in the impacted stream and *B. rhodani* and *A. standfussi* in the unimpacted stream (Fig. 4A, B). High scores on RDA 1

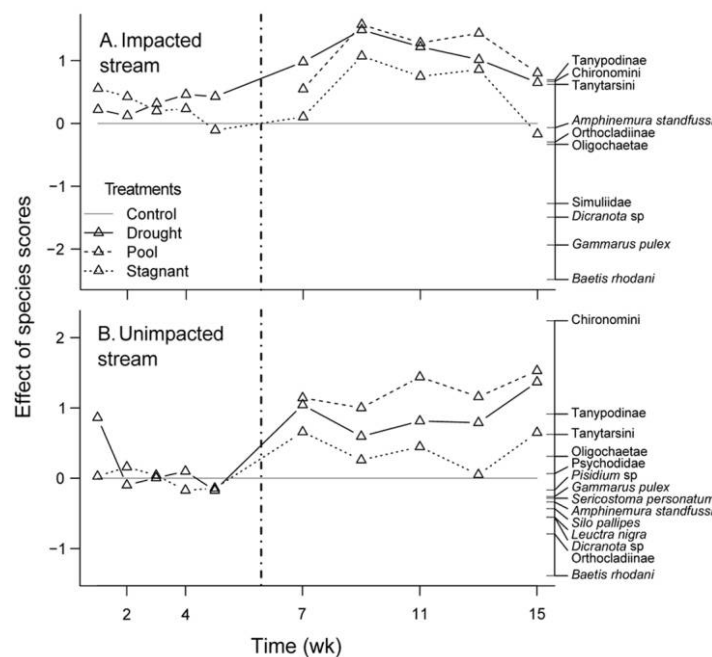


Figure 3. Principal response curves (PRCs) for macroinvertebrate abundance in the impacted (A) and unimpacted (B) stream. Each point represents the mean of 3 replicates per sampling point. The vertical broken line indicates the beginning of the experimental phase. See Fig. 1 for an explanation of PRCs.

Table 3. Mean (± 1 SE) treatment principal response curve (PRC) scores in impacted and unimpacted streams. ns = not significant.

Treatment	Impacted	Unimpacted
Drought	1.07 $\pm$ 0.32	0.92 $\pm$ 0.31 ^{ns}
Stagnant	0.52 $\pm$ 0.54	0.41 $\pm$ 0.26 ^{ns}
Pool	1.13 $\pm$ 0.42	1.26 $\pm$ 0.31 ^{ns}

were correlated with substrates consisting of leaves and organic mud and high coverage of submerged macrophytes in the impacted stream and with relatively high temperatures in the unimpacted stream. Chironomidae taxa were correlated with these environmental conditions in both streams (Fig. 4A, B). High scores on the 2nd d/b RDA axis (RDA 2) were correlated with substrates consisting of leaves and organic mud in the impacted stream and with high pH and substrates consisting of sand and silt in the unimpacted stream. None of the macroinvertebrate taxa were correlated with high RDA 2 scores in either stream. Low RDA 2 scores were not correlated significantly with any explanatory variable or macroinvertebrate taxon in either stream (Fig. 4A, B, Table 4).

DISCUSSION

Total abundance and diversity of macroinvertebrates did not respond negatively to experimentally induced flow reductions in the studied streams. This finding is contrary to previous observations (Englund and Malmqvist 1996, Dewson et al. 2007a) and may indicate that macroinvertebrate abundance and diversity do not respond ubiquitously to changes in flow conditions. Low flow can reduce total abundance and diversity of macroinvertebrates (Englund and Malmqvist 1996, Dewson et al. 2007a) by mediating reductions in O₂ availability at night, but increasing temperatures allow higher macroinvertebrate abundances (Rader and Belish 1999, our study) that may outweigh the effect of low O₂ concentrations. Thus, abundance and diversity measures are unlikely to capture changes in macroinvertebrate communities related to reduced flow adequately.

We observed a highly significant effect of reduced flow on macroinvertebrate community composition. These changes occurred within the first 2 wk after flow reductions and were driven primarily by changes in the abundance of *Baetis* sp. and Chironomidae in both streams and *G. pulex*, *Dicranota* sp., and Simuliidae in the impacted stream. Changes in community composition probably were directly coupled with treatment-induced changes in the physicochemical environment in the streams (i.e., changes in temperature, dissolved O₂, and flow) following reduction of discharge in the treated reaches. Water temperature increased in the treated reaches, reflecting the increased water residence, which allowed solar heating and reduced dissolved O₂ availability. Dissolved O₂ may have been further decreased by increased

microbial respiration in the treated stream reaches because we observed higher amounts of organic matter on the stream bed, probably because of higher retention (Dewson et al. 2007d) and reduced shredder abundance (e.g., *G. pulex*; Cuffney et al. 1990, Maltby et al. 2002) under reduced flow.

The decline in O₂ availability is the most likely explanation for the changes in the macroinvertebrate community

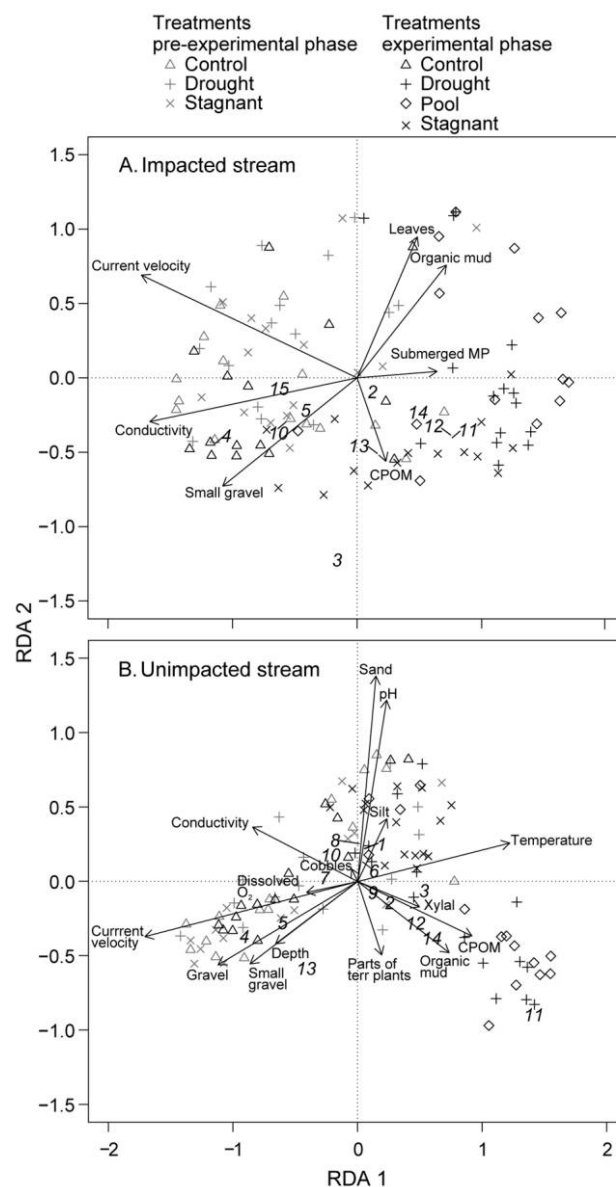


Figure 4. Distance-based redundancy analysis (d/b RDA) showing the relationship between faunistic and abiotic data in the impacted (A) and unimpacted (B) stream. Taxa shown are: 1 = *Pisidium* sp., 2 = *Oligochaeta*, 3 = *Gammarus pulex*, 4 = *Baetis rhodani*, 5 = *Amphinemura standfussi*, 6 = *Leuctra nigra*, 7 = *Silo pallipes*, 8 = *Sericostoma personatum*, 9 = Psychodidae, 10 = *Dicranota* sp., 11 = Chironominae, 12 = Tanytarsini, 13 = Orthocladiinae, 14 = Tanypodinae, 15 = Simuliidae. CPOM = coarse particulate organic matter, MP = macrophytes, terr = terrestrial.

Table 4. Results of permutational analysis of variance (1099 permutations) for explanatory variables for both streams. Some variables were collinear in the impacted stream and were excluded. CPOM = coarse particulate organic matter, MP = macrophyte. * indicates significance.

Stream	Explanatory variable	Variance	<i>F</i>	<i>P</i>
Impacted	Current velocity	2.7599	10.8528	0.0009*
	Conductivity	1.3148	5.1702	0.0009*
	Small gravel	1.1283	4.4370	0.0009*
	Organic mud	0.7557	2.9717	0.0009*
	Leaves	0.4914	1.9325	0.0109*
	Submerged MP	0.4232	1.6643	0.0364*
	CPOM	0.3965	1.5593	0.0791
	Residual	24.6674		
Unimpacted	Depth	1.3212	5.0190	0.0009*
	pH	0.8939	3.4298	0.0009*
	Conductivity	0.9507	3.6475	0.0009*
	Temperature	1.5577	5.9766	0.0009*
	Current velocity	1.7142	6.5771	0.0009*
	Gravel	1.0375	3.9806	0.0009*
	Small gravel	0.6838	2.6234	0.0036*
	Silt	0.4428	1.6989	0.0255*
	Dissolved O ₂	0.4135	1.5866	0.0491*
	Sand	0.4119	1.5803	0.0500
	Parts of terrestrial plants	0.2779	1.0661	0.3553
	Organic mud	0.3166	1.2148	0.1982
	Wood	0.2558	0.9814	0.4436
	Cobbles	0.2334	0.8954	0.5936
	CPOM	0.2279	0.8744	0.6391
	Residual	22.9358		

composition (Graeber et al. 2013), but the resolution of the available O₂ data was not sufficient to yield strong overall explanations in the RDA. We measured O₂ only at fixed points evenly distributed in the reach, and we did not measure habitat-specific O₂ concentrations. We also did not measure O₂ at night, and therefore, we did not capture the expected nighttime O₂ minima, although these probably are critical for the stream biota (Dewson et al. 2007a). We do not think that physicochemical variables sampled, but not included, in our study influenced the observed changes in the macroinvertebrate communities. Conductivity changes caused by treatments were within a range (50 μ S/cm in the unimpacted and 150 μ S/cm in the impacted stream) that is not considered to have biological influence on macroinvertebrates (Kefford 1998). pH fluctuations were minor, and pH remained neutral throughout the experimental phase. Therefore, the observed changes in pH probably did not influence the macroinvertebrate community because most macroinvertebrate species survive and reproduce normally between a pH 6.5 and 8.5 (Robertson-Bryan, Inc. 2004). Also 2 heavy precipitation events during the course of the experiment affected current

velocity of the stagnant reach of the impacted stream on 2 sampling occasions. Current velocity in the stagnant reach was >80% reduced compared to the control reach at the same time, so this increase probably did not influence the outcome of the study. Moreover, all treatments showed a significant change in macroinvertebrate community composition compared to the control. If anything, the changes in the stagnant reach of the impacted stream might have been more pronounced without these extreme events.

Several taxon-specific ecological adaptations might explain altered community composition under reduced flow. Reduced current velocity probably is a key factor for the lower occurrence of *Baetis* sp. and Simuliidae under low-flow conditions. *Baetis* sp. depends on flow for O₂ uptake and disappears in areas where current velocities drop well below their preference (0.9 m/s; Degani et al. 1993). Corrarino and Brusven (1983) reported similar results for Simuliidae, filter feeders that have very restricted tolerance to changes in water current and respond to unfavorable flow conditions by drifting. Reduced O₂ concentrations and increased temperatures probably were key factors explaining the lower occurrence of *G. pulex* under reduced-

flow conditions (Grant and Hawkes 1982). This species depends less than *Baetis* on a continuous flow for O₂ uptake because it can compensate with pleopod ventilation (Hughes 1970), but these movements lead to higher rates of respiration (Hughes 1970) and probably to lower survival rates. The mechanisms underlying the negative effect of low flow on *S. pallipes* and *Dicranota* sp. are less clear. The increase in abundance of Chironomidae corroborates previous findings that Chironomidae may be favored in drought conditions, probably because they are capable of exploiting food and habitats rapidly because of their high fecundity and short generation cycles (Ledger et al. 2011, 2012, 2013). One could argue that small taxa, such as Chironomidae, could be favored over larger taxa under drought conditions because they can better exploit the hyporheic zone, but we think it unlikely that the increase in the abundance of Chironomidae reflected a greater ability of this taxon to use the hyporheic zone as a refugium. No evidence exists, even in an intermittent stream, that macroinvertebrates use the hyporheic zone as a refugium during drying (del Rosario and Resh 2011).

Homogenization of stream habitats and habitat loss probably add to the effects of drought on macroinvertebrate communities (Cazaubon and Giudicelli 1999, Garcia-Roger et al. 2011, Walters and Post 2011). A more homogeneous stream environment developed in treated than in control reaches after flow reduction, and these changes probably contributed to changes in macroinvertebrate abundance. For example, *Baetis* sp. is negatively affected by soft sediments, low flow, and high water temperatures that, over time, became the prevailing conditions in the treated reaches (Rasmussen et al. 2012). In stagnant reaches, habitat area was preserved because the water body remained unchanged, but habitat area was reduced in drought reaches. Habitat loss may contribute to changes in abundance (Dewson et al. 2007a) and might explain the more pronounced effects in reaches with the most severe flow reductions (drought and pools).

Total abundance and diversity were similar in the experimentally created pools and the drought sections. Thus, we did not find evidence that the pools acted as a refugium for macroinvertebrates by retaining living space in the dry stream (Williams 1977, Delucchi 1989, James et al. 2008a). This finding may reflect the critically low dissolved O₂ concentrations in the experimental pools. These concentrations probably were lower than in the drought treatment, but we have no continuous O₂ data available to underpin this assertion. However, the observed accumulation of leaf litter, which was higher in pools than in the drought reaches, and high temperatures probably stimulated respiration with a concomitant reduction of O₂ availability, especially at night. Low dissolved O₂ concentration is typical of natural pools in intermittent streams (von Schiller et al.

2011), but such streams are likely to be inhabited by organisms able to tolerate low O₂ availability. In contrast, the streams we used normally have permanent water flow. Thus, species adapted to use of pools as refugia were not part of the existing communities. We might have obtained different results had we used intermittent streams with taxa that live in slow flowing or still water.

The observed overall response of the macroinvertebrate community to experimental flow reductions was similar between streams despite the fact that they differed widely in terms of nutrient availability and morphological features. This outcome differed from our expectation of less-distinct effects of flow reduction in the impacted stream, based on previous reports that macroinvertebrate communities in impacted streams are resilient to low-flow-induced changes in the physicochemical environment (Dewson et al. 2007b, c, Death et al. 2009). Our results might indicate that macroinvertebrate community structure and ecosystem functions were closely connected in the streams. Higher nutrient availability together with higher temperatures in the impacted stream may have stimulated microbial activity (e.g., denitrification; Inwood et al. 2005) relative to the unimpacted stream, and therefore, the physicochemical changes may have been more drastic. Low-flow-induced changes in physicochemical conditions may have exacerbated the response in the impacted stream, outweighing the expected higher resilience of this stream's community. We also think it likely that flow reductions in the stagnant and drought reaches far exceeded the possible adaptations of the existing macroinvertebrates because neither of the existing stream communities was preadapted to stagnant or drought conditions.

CONCLUSIONS

Ours is one of the first field experiments done to investigate macroinvertebrate community responses to extreme flow reduction. We clearly demonstrated that stagnant and drought conditions subsequent to altered flow significantly alter macroinvertebrate community composition and that these changes probably are closely connected to physicochemical changes in the environment and to ecosystem functions (e.g., organic matter decomposition and nutrient cycling). Pools did not act as substantial refugia for macroinvertebrates in our study.

Changes in community composition occurred immediately after flow reduction and were related primarily to changes in composition of the macroinvertebrate community rather than to changes in diversity. Therefore, diversity measures seem unsuited to detect short-term effects of stagnation and drought on stream macroinvertebrate communities. We found clear indications that community changes were related to increased temperatures, higher amounts

of organic matter, changes in current velocity, and a decline in O₂ availability. Moreover, we may have underestimated the negative effect of reduced O₂ availability on the macroinvertebrate community because we did not measure nighttime minima. Macroinvertebrate responses to flow reduction were independent of stream nutrient level, probably because the physicochemical changes were exacerbated in the impacted stream and outweighed the expected higher resilience of this community. We recommend further research using whole-community approaches in natural intermittent systems to evaluate the influence of pools on the stability of these systems. Emphasis should be put on life-history traits of species that can withstand sudden drying to identify adaptations necessary to survive climate-induced changes in streams.

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